

Model Identification and Data Analysis 1

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Introduction

These notes are based on the lectures delivered by Professor Simone Formentin during the Model Identification and Data Analysis 1 course at Politecnico di Milano during Semester 2025-1. Official course information can be found on the official program (Formentin, 2025). Some exercise sessions, were taught by Teaching Assistant Marco Rapizza.

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1 Introduction

1.1 Definitions

Definition 1.1: System S

Physical mechanism transforming inputs into outputs

Definition 1.2: Model M

Mathematical description of a system

Examples of systems:

- Economical: Relationship between incomes and education
- Social: Relationship between crime rates and

Three approaches to modeling:

1. White-box modeling: Based on physical laws or prior knowledge.

The pros are that all the variables have a physical meaning and the model could be generalized. On the other hand, the cons are that everything must be known a priori, it is expensive and time-consuming, so it is not feasible with complex systems.

2. Black-box modeling: Based on experimental data, it estimates the parameters based on statistical relationships. We do not know what's happening inside the model, we only know its inputs and outputs.

The pros are that it can save a lot of computational resources and time, and we don't need a deep process model. The cons are that the model is not physically interpretable, and if the system changes, we must repeat the experiments.

3. Gray-box Model: A mid-way model.

This course is about black-box modeling.

The model quality is assessed by comparing the real and simulated output. If the outputs are similar, according to measures to be defined, the model accurately replicates the system dynamics.

To build a system model, we can use data and prior knowledge of a model.

1.2 Static Systems

A static system is a system where the knowledge of the inputs variables is enough to estimate the output.

1.3 Dynamical Systems

A dynamical system is a system where the past behavior of the output influences the value of the output in the present. It is needed knowledge of the current status of the system.

The time dimension becomes critical, and it can be continuous or discrete. Natural and physical phenomena are intrinsically continuous. We can use mathematical tools for modeling. Ordinary Differential Equations are the most used ones, since the derivative gives us a notion of the function behavior in the future. Anyway, computers can only compute discrete values, so a finite set of points can be calculated, usually, these points are equally distributed.

2 Stochastic Processes

2.1 Definitions

Definition 2.1: Stochastic Process (SP)

Is an infinite sequence of random variables, all defined on the same probabilistic space

The distribution can change through the time

It is notated as:

$$v(1, s), v(2, s), \dots, v(t, s), \dots$$

SP extends the notion of a random variable to signals

We will notate $v(t, \bar{s})$ when we have a deterministic signal.

We will notate $v(\bar{t}, s)$ for a specific random variable.

Given 3 different signals $v_1(t), v_2(t), v_3(t)$. Even if they are different realizations of the same SP $v(t, s)$, they are said to be stochastically equivalent.

To fully describe a SP, we need to know how the probability distribution or $v(s)$ evolves through the time t .

2.2 Mean and Covariance

From now on, we will work with the so-called wide-sense characterization of SP: Mean + Covariance function

Definition 2.2: Mean value of a SP

$$m(t) = \mathbb{E}(v(t, s)) = \int_{\mathbb{R}} v(t, s) pdf(s) ds \quad (\text{pdf}=\text{density function})$$

Definition 2.3: Covariance function of a SP

$$\gamma(t_1, t_2) = \mathbb{E}[(v(t_1, s) - m(t_1))(v(t_2, x) - m(t_2))]$$

If $t_1 = t_2$:

$$\gamma(t_1, t_1) = \mathbb{E}((v(t_1) - m(t_1))^2)$$

2.3 Stationary Process

Definition 2.4: Stationary Stochastic Process

It is an SP with:

1. $m(t) = m \quad \forall t$
2. $\gamma(t_1, t_2) = \gamma(t_1, t_1 + \tau) = \gamma(\tau)$

The second property implies that

$$\gamma(t_1, t_2) = \gamma(t_3, t_4) \quad \forall t_1, t_2, t_3, t_4 \quad \text{st} \quad t_4 - t_3 = t_2 - t_1$$

Properties of $\gamma(\tau)$:

- Positivity: $\gamma(0) = \mathbb{E}[(v(t, s) - m)^2] \geq 0$
- Non-increasing: $|\gamma(\tau)| \leq \gamma(0) \quad \forall \tau$
- Even function: $\gamma(\tau) = \gamma(-\tau) \quad \forall \tau$

Observations:

- Given a SSP $v(t, s)$, we will write $m_v(t)$ (or m_v) and $\gamma_v(t_1, t_2)$ (or $\gamma_v(\tau)$) to indicate its mean and covariance function.
- Two SSPs $v_1(t, s)$ and $v_2(t, s)$ are wide-sense equivalent if $m_1 = m_2$ and $\gamma_{v_1}(\tau) = \gamma_{v_2}(\tau) \quad \forall \tau$
- The covariance function $\mathbb{E}[(v(t_1, s) - m)(v(t_2, s) - m)]$ is different from the correlation function $E[v(t_1, s) \cdot v(t_2, s)]$ if $m \neq 0$

2.4 Exercises

Exercise 1: Let be $v(t, s) = \alpha(s)$, $\alpha(s) \sim N(1, 3) = N(\mu, \sigma^2)$. Is this process stationary?

First condition:

$$m(t) = \mathbb{E}(v(t, s)) = \mathbb{E}(\alpha(s)) = 1 \quad \forall t$$

Second condition:

$$\alpha(t_1, t_2) = \mathbb{E}((v(t_1, s) - 1)(v(t_2, s) - 1)) = \mathbb{E}((\alpha(s) - 1)^2) = 3 \quad \forall t \forall \tau$$

Then it is stationary

Exercise 2: Let be $v(t, s) = t \cdot \alpha(s) - t$, $\alpha(s) \sim N(1, 3) = N(\mu, \sigma^2)$. Is this process stationary?

First condition:

$$m(t) = \mathbb{E}(v(t, s)) = \mathbb{E}(t \cdot \alpha(s) - t) = t\mathbb{E}(\alpha(s)) - \mathbb{E}(t) = 0 \quad \forall t$$

Second condition:

$$\begin{aligned} \alpha(t_1, t_2) &= \mathbb{E}[(v(t_1, s) - 0)(v(t_2, s) - 0)] = \mathbb{E}[v(t_1, s) \cdot v(t_2, s)] \\ &= \mathbb{E}[(t \cdot \alpha(s) - t)((t + \tau) \cdot \alpha(s) - (t + \tau))] \quad \forall t \forall \tau \\ &= \mathbb{E}[(t\alpha(s) - t)(t\alpha(s) + \tau\alpha(s) - t - \tau)] \quad \forall t \forall \tau \\ &= \mathbb{E}[t^2\alpha^2(s) + t\tau\alpha^2(s) - t^2\alpha(s) - t\tau\alpha(s) - t^2\alpha(s) - t\tau\alpha(s) + t^2 + t\tau] \quad \forall t \forall \tau \\ &= t^2\sigma^2 + t\tau\sigma^2 + t^2 + t\tau = (t^2 + t\tau)(1 + \sigma^2) \end{aligned}$$

Then it is not stationary

2.5 White Noise

Definition 2.5: White noise

A SSP with mean μ and variance λ^2 is called white noise (WN) if the following properties hold:

1. $\mathbb{E}[e(t)] = \mu \quad \forall t$
2. $\mathbb{E}[(e(t) - \mu)^2] = \lambda^2 \quad \forall t$
3. $\gamma_e(\tau) = \mathbb{E}[(e(t) - \mu)(e(t + \tau) - \mu)] = 0 \quad \forall t \forall \tau \neq 0$

We write $e(t) \sim WN(\mu, \lambda^2)$.

The constant realization is admissible but it is very unlikely.

3 Moving Averages

3.1 Finite Moving Averages

Definition 3.1: Finite Moving Averages

Let $e(t) \sim WN(0, \lambda^2)$. Define the Finite Moving Average Process of order m :

$$y(t) = c_0 \cdot e(t) + c_1 \cdot e(t - 1) + c_2 \cdot e(t - 2) + \dots + c_m \cdot e(t - m)$$

It is denoted as $MA(m)$

3.2 Stationarity

$MA(m)$ is a stationary stochastic process:

- The mean is constant

$$\begin{aligned}
m_y(t) &= \mathbb{E}[y(t)] = \mathbb{E}[c_0 e(t) + c_1 e(t-1) + \cdots + c_m e(t-m)] \\
&= c_0 \mathbb{E}[e(t)] + c_1 \mathbb{E}[e(t-1)] + \cdots + c_m \mathbb{E}[e(t-m)] \\
&= \mathbb{E}[e(t)] \sum_{i=1}^m c_i = 0 \quad \forall t \quad (\text{Since } \mathbb{E}[e(t)] = 0)
\end{aligned}$$

- Independency on t_1, t_2 for γ .

For $\tau = 0$

$$\begin{aligned}
\gamma_y(t, t) &= \mathbb{E}[(y(t) - m_y)^2] = \mathbb{E}[y(t)^2] \\
&= \mathbb{E}[(c_0 \cdot e(t) + c_1 \cdot e(t-1) + c_2 \cdot e(t-2) + \cdots + c_m \cdot e(t-m))^2] \\
&= \mathbb{E}[c_0^2 \cdot e(t)^2 + c_1^2 \cdot e(t-1)^2 + c_2^2 \cdot e(t-2)^2 + \cdots + c_m^2 \cdot e(t-m)^2 \\
&\quad + 2c_0 c_1 e(t) e(t-1) + 2c_0 c_2 e(t) e(t-2) + \cdots + 2c_{m-1} c_m e(t-m+1) e(t-m)] \\
&= c_0^2 \cdot \mathbb{E}[e(t)^2] + c_1^2 \cdot \mathbb{E}[e(t-1)^2] + c_2^2 \cdot \mathbb{E}[e(t-2)^2] + \cdots + c_m^2 \cdot \mathbb{E}[e(t-m)^2] \\
&\quad + 2c_0 c_1 \cdot \mathbb{E}[e(t) e(t-1)] + 2c_0 c_2 \cdot \mathbb{E}[e(t) e(t-2)] + \cdots + 2c_{m-1} c_m \cdot \mathbb{E}[e(t-m+1) e(t-m)]
\end{aligned}$$

Now, using the properties of WN, and $\mu = 0$

$$\begin{aligned}
\gamma_y(t, t) &= c_0^2 \cdot \lambda^2 + c_1^2 \cdot \lambda^2 + \cdots + c_m^2 \cdot \lambda^2 + 0 + \cdots + 0 \\
&= \lambda^2 \sum_{i=1}^m c_i^2 \quad \forall t
\end{aligned}$$

Now use $\tau = 1$

$$\begin{aligned}
\gamma_y(t, t-1) &= \mathbb{E}[(y(t) - m_y)(y(t-1) - m_y)] = \mathbb{E}[y(t)y(t-1)] \\
&= \mathbb{E}[c_0^2 e(t)e(t-1) + c_0 c_1 e(t)e(t-2) + \cdots + c_0 c_m e(t)e(t-m-1) + \\
&\quad + c_1 c_0 e(t-1)^2 + c_1^2 e(t-1)e(t-2) + \cdots + c_m^2 e(t-m)e(t-m-1)] \\
&= \lambda^2 \sum_{i=1}^m c_i c_{i-1} \quad \forall t
\end{aligned}$$

We can prove the calculation does not depends on t_1, t_2 either for larger τ . For instance, for $\tau = 2$

$$\gamma_y(t, t-1) = \lambda^2 \sum_{i=2}^m c_i c_{i-2} \quad \forall t$$

Eventually, for $\tau = m$

$$\gamma_y(t, t-m) = \lambda^2 c_0 c_m \quad \forall t$$

and, for $\tau > m$

$$\gamma_y(t, t-m-1) = 0 \quad \forall t$$

So we prove the independency over t_1, t_2 for every τ

Remark.

We say the process is “colored” for the τ s where there is a covariance different from 0 between t_1 and $t_1 + \tau$

3.3 Infinite Moving Averages

Definition 3.2: Infinite Moving Averages

Let $e(t) \sim WN(0, \lambda^2)$. Assume that $\sum_{i=0}^{+\infty} c_i^2 < +\infty$. Define the Infinite Moving Average Process:

$$y(t) = \sum_{i=0}^{+\infty} c_i e_{t-i}$$

It is denoted as $MA(\infty)$

The mean is:

$$m_y(t) = \mathbb{E}[y(t)] = \mathbb{E}\left[\sum_{i=0}^{\infty} c_i e(t-i)\right] = \sum_{i=0}^{\infty} c_i \mathbb{E}[e(t-i)] = 0$$

The variance is:

$$\begin{aligned} \gamma_y(t, t) &= \mathbb{E}[y(t)^2] = \mathbb{E}\left[\left(\sum_{i=0}^{+\infty} c_i e(t-i)\right)\left(\sum_{j=0}^{+\infty} c_j e(t-j)\right)\right] \\ &= \mathbb{E}\left[\sum_{i=0, j=0}^{\infty} c_i c_j e(t-i)e(t-j)\right] = \sum_{i=0, j=0}^{\infty} c_i c_j \mathbb{E}[e(t-i)e(t-j)] \\ &= \sum_{i=0}^{\infty} c_i^2 \lambda^2 < \infty \end{aligned}$$

The covariance is:

$$\begin{aligned} \gamma_y(t, t-\tau) &= \mathbb{E}\left[\left(\sum_{i=0}^{+\infty} c_i e(t-i)\right)\left(\sum_{j=0}^{+\infty} c_j e(t-j-\tau)\right)\right] \\ &= \mathbb{E}\left[\sum_{i=0, j=0}^{\infty} c_i c_j e(t-i)e(t-j-\tau)\right] \\ &= \sum_{i=0, j=0}^{\infty} c_i c_j \mathbb{E}[e(t-i)e(t-j-\tau)] \\ &= \lambda^2 \sum_{i=0}^{\infty} c_i c_{i+\tau} \end{aligned}$$

Then we can check that the infinite moving average is also a stationary process.

In this case, $MA(\infty)$ is an SSP that can model any SSP process

4 Autoregressive Models

4.1 AR(M)

MA uses the function of $\gamma(\tau)$ to model the system, where τ is often referred to as ‘lag’. There exists a class of SSP models (Even the infinite ones, as $MA(\infty)$) that can shape $\gamma(\tau)$ for all τ s with a finite set of coefficients.

Definition 4.1: AR(m)

Let $e(t) \sim WN(0, \lambda^2)$ and consider:

$$y(t) = a_1 y(t-1) + a_2 y(t-2) + \dots + a_m y(t-m) + e(t)$$

Then $y(t)$ is called Autoregressive Model of order m , and it is denoted as $AR(m)$.

AR is the steady-state solution of a differential equation. A steady-state solution of a differential equation is a solution that does not care about the border conditions.

Example:

$$\begin{aligned} y(t) &= ay(t-1) + e(t) \\ &= a(ay(t-2) + e(t-1)) + e(t) = a^2 y(t-2) + ae(t-1) + e(t) \\ &= a^2(ay(t-3) + e(t-2)) + ae(t-1) + e(t) = a^3 y(t-3) + a^2 e(t-2) + ae(t-1) + e(t) \\ &\dots \\ &= \underbrace{e(t) + ae(t-1) + a^2 e(t-2) + \dots}_{\text{External Excitation}} + \underbrace{a^{t-t_0} y(t_0)}_{\text{Initial Condition}} \end{aligned}$$

AR does not consider the initial condition part. Actually, if we made a model considering the initial condition, it wouldn’t be stationary, but AR is stationary.

We can note, that we can express the $MA(\infty)$ with $c_i = a^i$, with an $AR(1)$. Anyway, power of $MA(\infty)$ is still higher of $AR(1)$, as the coefficients are constrained.

4.2 ARMA

Definition 4.2: ARMA

Let be $e(t) \sim WN(0, \lambda^2)$ and consider:

$$y(t) = a_1 y(t-1) + a_2 y(t-2) + \dots + a_m y(t-m) + c_0 e(t) + c_1 e(t-1) + \dots + a_n y(t-n)$$

This is called an $ARMA(m,n)$

In practice, we use this model to express the stochasticity of variables not accounted for in the input we are taking.

4.3 ARMAX

Definition 4.3: ARMAX

Let be $e(t) \sim WN(0, \lambda^2)$ and consider:

$$\begin{aligned} y(t) = & a_1 y(t-1) + a_2 y(t-2) + \dots + a_m y(t-m) + \\ & c_0 e(t) + c_1 e(t-1) + \dots + c_n e(t-n) + \\ & b_0 u(t-k) + b_1 u(t-k-1) + \dots + b_p u(t-k-p) \end{aligned}$$

This is called an ARMAX(m,n,k,p).

The X stands for exogenous.

The exogenous part takes a delay of k steps to start to act upon a difference in the input.

Remark.

■ We will assume that u is deterministic and fully known.

4.4 N-ARMAX

Definition 4.4: N-ARMAX

Let be $e(t) \sim WN(0, \lambda^2)$ and consider:

$$\begin{aligned} y(t) = & f(y(t-1), y(t-2), \dots, y(t-m), \\ & e(t), e(t-1), \dots, e(t-n), \\ & u(t-k), u(t-k-1), \dots, u(t-k-p)) \end{aligned}$$

f could be any non-linear parametric function, like polynomials, wavelets, splines, neural networks, etc.

This is called an N-ARMAX(m,n,k,p)

The N stands for non-linear.

4.5 Operatorial Representation

We need additional tools to study the stationarity of AR models.

Definition 4.5: backward-shift operator

The backward-shift operator z^{-1} is defined as:

$$z^{-1}(x(t)) = x(t-1)$$

Definition 4.6: forward-shift operator

The forward-shift operator z^{+1} is defined as:

$$z^{+1}(x(t)) = x(t+1)$$

Properties of z^{-1} and z^{+1} :

- Linearity:

$$z^{-1}(ax(t) + by(t)) = ax(t-1) + by(t-1)$$

- Recursive Application:

$$z^{-1}(z^{-1}(z^{-1}(x(t)))) = z^{-1}(z^{-1}(x(t-1))) = z^{-1}(x(t-2)) = x(t-3) = z^{-3}x(t)$$

- Linear Composition:

$$(az^{-1} + bz + cz^{-3} + dz^{+2})x(t) = ax(t-1) + bx(t+1) + cx(t-3) + dx(t+2)$$

4.6 Transfer Functions

Given an ARMA(m,n) model:

$$y(t) = a_1y(t-1) + a_2y(t-2) + \dots + c_0e(t) + \dots + c_n e(t-n)$$

We can rewrite it:

$$y(t) = a_1z^{-1}y(t) + a_2z^{-2}y(t) + \dots + c_0e(t) + \dots + c_nz^{-n}e(t)$$

Then:

$$(1 - a_1z^{-1} - a_2z^{-2} - \dots - a_mz^{-m})y(t) = (c_0 + c_1z^{-1} + c_2z^{-2} + \dots + c_nz^{-n})e(t)$$

Therefore:

$$y(t) = \frac{(c_0 + c_1z^{-1} + c_2z^{-2} + \dots + c_nz^{-n})}{(1 - a_1z^{-1} - a_2z^{-2} - \dots - a_mz^{-m})}e(t) = \frac{C(z)}{A(z)}e(t) = W(z)e(t)$$

Definition 4.7: Transfer Function

$W(z)$ in the last expression is called the discrete-time transfer function

This representation of the model is useful when we want to analyze whether the model is stationary or not.

$$W(z) = \frac{C(z)}{A(z)}$$

Definition 4.8: Zeros

The roots of $C(z)$ in $W(z)$ are called zeros.

Definition 4.9: Zeros

The roots of $A(z)$ in $W(z)$ are called poles.

Remark.

- MA(n) has n non-trivial zeros and n poles in the origin. Therefore, the process is often called as ‘all-zero’ process.
- AR(m) has m non-trivial poles and m zeros in the origin. Therefore, the process is often called as ‘all-poles’ process.

4.7 Operations with Transfer Functions

Let $W(z)$ and $M(z)$ be transfer functions of linear digital filters where:

$$y(t) = M(W(u(t)))$$

We notate:

$$\begin{aligned} y(t) &= M(z) \cdot s(t) \\ s(t) &= W(z) \cdot u(t) \end{aligned}$$

On the other hand, we can also have parallel interconnection between transfer functions:

$$y(t) = x(t) + v(t)$$

Where:

$$\begin{aligned} x(t) &= W(z)u(t) \\ v(t) &= M(z)u(t) \end{aligned}$$

So $y(t) = W(z)u(t) + M(z)u(t)$

4.8 Transfer Functions in ARMAX

Particularly, in ARMAX models:

$$\begin{aligned}
 y(t) &= a_1 y(t-1) + a_2 y(t-2) + \cdots + a_m y(t-m) + \\
 &\quad c_0 e(t) + c_1 e(t-1) + \cdots + c_n e(t-n) + \\
 &\quad b_0 u(t-k) + b_1 u(t-k-1) + \cdots + b_p u(t-k-p) \\
 &= a_1 z^{-1} y(t) + a_2 z^{-2} y(t) + \cdots + a_m z^{-m} y(t) + \\
 &\quad c_0 e(t) + c_1 z^{-1} e(t) + \cdots + c_n z^{-n} e(t) + \\
 &\quad b_0 z^{-k} u(t) + b_1 z^{-k} z^{-1} u(t) + \cdots + b_p z^{-k} z^{-p} u(t)
 \end{aligned}$$

Then:

$$\underbrace{(1 - a_1 z^{-1} - a_2 z^{-2} - \cdots - a_m z^{-m})}_{A(z)} y(t) = \underbrace{(c_0 + c_1 z^{-1} + c_2 z^{-2} + \cdots + c_n z^{-n})}_{C(z)} e(t) + \underbrace{(b_0 + b_1 z^{-1} + b_p z^{-2} + \cdots + b_p z^{-p})}_{B(z)} z^{-k} u(t)$$

Therefore:

$$y(t) = \frac{C(z)}{A(z)} e(t) + \frac{B(z)}{A(z)} z^{-k} u(t) \quad e(t) \sim WN(0, \lambda^2)$$

Remember that $u(t)$ is real, whereas $e(t)$ is fictitious (it is a dependent variable).

Remark.

Warning: These are equivalent:

$$\frac{c_0 + c_1 z^{-1}}{1 - a z^{-1}} = \frac{c_0 z + c_1}{z - a}$$

Both negative and positive powers of z can be used.

Be careful with mixed expressions, this is incorrect: $\frac{c_0 z}{1 + a z^{-1}}$.

4.9 Stationarity with $W(z)$

Definition 4.10: Asymptotically Stable

A linear digital filter with transfer function $W(z)$ is asymptotically stable $\iff$:

all its poles are strictly inside the unit circle in the complex plane

If it is not, the system is said to be unstable.

Remark.

When the zeros are in the unit circle, the process is said to be minimum phase.

Theorem 4.11:

The steady-state output $y(t) = W(e(t))$ is stationary if and only if:

- $e(t)$ is SSP
- $W(z)$ is asymptotically stable

4.10 Mean and Covariance

We can calculate the mean and covariance of stationary processes only. So the calculation is done assuming always stationarity.

Let's start from a simple case:

$$AR(1) : y(t) = ay(t-1) + e(t), \quad e(t) \sim WN(0, \lambda^2)$$

We can represent it as:

$$y(t) = W(z)e(t) = \frac{z}{z-a}e(t) = \frac{1}{1-az^{-1}}e(t)$$

We could represent it as $MA(\infty)$ and use the tools already known to compute $E(y(t))$ and $\gamma(\tau)$, but that's not possible due to the number of terms.

We can, instead, use the properties of expectation to calculate the mean and covariance as unknown variables, using the model's stationarity, which implies that the expectation is the same at any point.

Proposition 4.12

For any AR process:

$$E[e(t)y(t-\tau)] = 0 \quad \forall \tau \geq 0$$

Proof. Take the case of $AR(1)$, it's extrapolable to general p.

Since:

$$y(t) = e(t) + ae(t-1) + a^2e(t-2) + \dots$$

Then:

$$y(t-1) = e(t-1) + ae(t-2) + a^2e(t-3) + \dots$$

So:

$$\begin{aligned} E[e(t)y(t-1)] &= E[e(t) \cdot (e(t-1) + ae(t-2) + a^2e(t-3) + \dots)] \\ &= E[e(t)e(t-1)] + aE[e(t)e(t-2)] + \dots \\ &= 0 \quad (\text{Since iid of } e) \end{aligned}$$

□

Take now the case of ARMA(m,n):

$$y(t) = a_1 y(t-1) + \dots + a_m y(t-m) + c_0 e(t) + \dots + c_n e(t-n)$$

Assume $a_1, \dots, a_m$ are such that $y(t)$ is SSP.

Mean:

$$\begin{aligned} E[y(t)] &= m_y = E[a_1 y(t-1) + \dots + a_m y(t-m) + c_0 e(t) + \dots + c_n e(t-n)] \\ &= a_1 \underbrace{E[y(t-1)]}_{m_y} + \dots + a_m \underbrace{E[y(t-m)]}_{m_y} + c_0 \underbrace{E[e(t)]}_{m_e} + \dots + c_n \underbrace{E[e(t-n)]}_{m_e} \end{aligned}$$

Implies:

$$\begin{aligned} (1 - a_1 - a_2 - \dots - a_m)m_y &= (c_0 + c_1 + \dots + c_m)m_e \\ \therefore m_y &= \frac{c_0 + c_1 + \dots + c_m}{1 - a_1 - a_2 - \dots - a_m} m_e = 0 \quad (\text{Since } m_e \text{ is } 0) \end{aligned}$$

Covariance:

$$\begin{aligned} \gamma_y(0) &= E[y(t)^2] = E[(a_1 y(t-1) + \dots + a_m y(t-m) + c_0 e(t) + \dots + c_n e(t-n))^2] \\ &= E[a_1^2 y(t-1)^2 + \dots + c_n^2 e(t-n)^2 + \\ &\quad 2a_1 a_2 y(t-1) \cdot y(t-2) + \dots \\ &\quad 2a_1 c_0 e(t) \cdot y(t-1) + \dots \\ &\quad 2c_0 c_1 e(t)e(t-1) + \dots] \end{aligned}$$

Following this process for $\tau = 0, 1, 2, \dots$:

$$\begin{cases} \gamma_y(0) = a_1^2 \gamma_y(0) + a_2^2 \gamma_y(0) + \dots + 2a_1 a_2 \gamma_y(1) + \dots \\ \gamma_y(1) = a_1 \gamma_y(0) + \dots + c_1 e \cdot \gamma_1(1) + \dots + \gamma_y(2) + \dots \\ \dots \\ \gamma_y(m-1) = a_1 \gamma_y(m-2) + \dots \end{cases}$$

Having a set of m recursive equations called the Yule-Walker Equations for ARMA(m,n).

4.11 Mean and Covariance with non-zero mean

Despite there's always possible to calculate the mean and variance as indicated in section 4.10, it could be much more complicated if we have a non-zero mean process. That's why we often detrend the processes before calculating these metrics:

Definition 4.13: Debias

Let be $y(t)$ a stationary sthochastic process and $e(t)$ a WN process. Define $\tilde{y}$ and $\tilde{e}(t)$ as:

$$\begin{cases} \tilde{y}(t) = y(t) - m_y \\ \tilde{e}(t) = e(t) - y \end{cases}$$

Remark.

$\tilde{y}$ and $\tilde{e}(t)$ are unbiased versions of $e(t)$ and $y(t)$.
Let be $e(t) \sim WN(\mu, \lambda^2)$:

$$E(e(t)) = \mu$$

For $\tau = 0$:

$$\begin{aligned}\lambda^2 &= E[(e(t) - \mu)^2] = E[(e(t)^2 + \mu^2 - 2\mu \cdot e(t))] \\ &= E[e(t)^2] + E[\mu^2] - 2E[\mu \cdot e(t)] \\ &= E[e(t)^2] + \mu^2 - 2\mu \cdot \mu = E[e(t)^2] - \mu^2 \\ \implies E[e(t)^2] &= \lambda^2 + \mu^2\end{aligned}$$

For $\tau \neq 0$:

$$\begin{aligned}0 &= E[(e(t) - \mu)(e(t - \tau) - \mu)] \\ &= E[(e(t)e(t - \tau)) + E[\mu^2] - \mu \cdot E[e(t - \tau)] - \mu \cdot E[e(t)]] \\ &= E[(e(t)e(t - \tau)) + E[\mu^2] - \mu \cdot \mu - \mu \cdot \mu] \\ \implies E[e(t)e(t - \tau)] &= \mu^2\end{aligned}$$

On the other hand, write $\tilde{y}(t)$ as function of past $\tilde{y}(t)$:

$$\begin{aligned}\tilde{y}(t) &= y(t) - m_y \\ &= a_1 \cdot y(t - 1) + a_2 \cdot y(t - 2) + \dots + a_m \cdot y(t - m) \\ &\quad + c_0 \cdot e(t) + c_1 \cdot e(t - 1) + \dots + c_n \cdot e(t - n) - m_y \\ &= a_1 \cdot (\tilde{y}(t - 1) + m_y) + a_2 \cdot (\tilde{y}(t - 2) + m_y) + \dots + a_m \cdot (\tilde{y}(t - m) + m_y) \\ &\quad + c_0 \cdot (\tilde{e}(t) + \mu) + c_1 \cdot (\tilde{e}(t - 1) + \mu) + \dots + c_n \cdot (\tilde{e}(t - n) + \mu) - m_y \\ &= a_1 \cdot \tilde{y}(t - 1) + \dots + a_m \cdot \tilde{y}(t - m) + c_0 \cdot \tilde{e}(t) + \dots + c_n \cdot \tilde{e}(t - n) \\ &\quad + m_y \left(\sum_{i=1}^m a_i - 1 \right) + \mu \sum_{i=0}^n c_i\end{aligned}$$

Remember that:

$$m_y = \frac{c_0 + c_1 + \dots + c_m}{1 - a_1 - a_2 - \dots - a_m} \mu$$

The terms above are canceled:

$$\tilde{y}(t) = a_1 \cdot \tilde{y}(t - 1) + \dots + a_m \cdot \tilde{y}(t - m) + c_0 \cdot \tilde{e}(t) + \dots + c_n \cdot \tilde{e}(t - n)$$

So, we can calculate easily $\gamma_y(\tau)$ by calculating $\gamma_{\tilde{y}}(\tau)$ as the covariance function of a process generated by a zero-mean WN $(\tilde{e}(t))_t$:

$$\gamma_{\tilde{y}}(\tau) = E[\tilde{y}(t)\tilde{y}(t - \tau)] = E[(y(t) - m_y)(y(t - \tau) - m_y)] = \gamma_y(\tau)$$

5 Analysis in the Frequency Domain

5.1 Spectrum

Definition 5.1: Spectrum

The spectral density (or “spectrum”) of a SSP $y(t)$ is:

$$\Gamma_y(\omega) = \sum_{\tau=-\infty}^{\infty} \gamma_y(\tau) e^{-j\omega\tau}$$

Remark.

$\Gamma_y(\omega) \equiv F\{\gamma_y(\tau)\}$ is the discrete Fourier transform (DFT) of the covariance function $\gamma_y(\tau)$.

Properties:

- $\gamma_y(\omega)$ is a real function of a real variable ω , so:

$$\text{Im}(\Gamma_y(\omega)) = 0 \quad \forall \omega \in \mathbb{R}$$

- Positive:

$$\Gamma_y(\omega) \geq 0 \quad \forall \omega \in \mathbb{R}$$

- Even:

$$\Gamma_y(\omega) = \Gamma_y(-\omega) \quad \forall \omega \in \mathbb{R}$$

- 2π -periodic:

$$\Gamma_y(\omega) = \Gamma_y(\omega + 2\pi k) \quad \forall \omega \in \mathbb{R} \quad \forall k \in \mathbb{Z}$$

Remark.

The minimum period that can be represented is 2 samples.

Actually, if $\omega \leq \pi = \omega_{\max}$ (nyquist frequency)

$$\begin{aligned} \frac{2\pi}{T} &= \omega \leq \pi = \omega_{\max} \\ T &\geq T_{\min} = \frac{2\pi}{\omega_{\max}} = \frac{2\pi}{\pi} = 2 \end{aligned}$$

Definition 5.2: Inverse Transform

The inverse transform of $\Gamma_y(\omega)$ is the covariance term on τ

$$\gamma_y(\tau) = F^{-1}\{\Gamma_y(\omega)\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma_y(\omega) e^{j\omega\tau} d\omega$$

$\gamma_y(\tau)$ and $\Gamma_y(\omega)$ carry the same information about $y(t)$

Remark.

■

For $\tau = 0$:

$$\gamma_y(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma_y(\omega) d\omega$$

■

Note that this integral is the area under $\Gamma_y(\omega)$. So only with a representation,

■

having the area of the function, we can calculate the variance of the process

5.2 White noise

The spectrum of a White noise is:

$$\begin{aligned} \Gamma_e(\omega) &= \sum_{\tau=-\infty}^{\infty} \gamma_e(\tau) e^{-j\omega\tau} \\ &= \gamma_e(0) \cdot e^{-j\omega 0} + \gamma_e(1) \cdot e^{-j\omega 1} + \gamma_e(-1) \cdot e^{+j\omega 1} + \gamma_e(2) \cdot e^{-j\omega 2} + \gamma_e(-2) \cdot e^{+j\omega 2} + \dots \\ &= \gamma_e(0) \cdot e^{-j\omega 0} \quad \text{For WN, } \gamma_e(\tau) \text{ is 0 for } \tau \neq 0 \\ &= \gamma_e(0) \cdot 1 = \lambda^2 \end{aligned}$$

So the spectrum is a constant equal to λ^2 between $-\pi$ to π . In this case, we say that the process is flat.

5.3 ARMA

We cannot use the definition to compute $\Gamma_y(\omega)$ for a generic ARMA $y(t)$, since we would have infinite $\gamma_y(\tau)$. In this case, we can use the next theorem.

Theorem 5.3: Fundamental Theorem of the Spectral Analysis

Let be $y(t) = W(v(t))$. If $v(t)$ is SSP and $W(z)$ is asymptotically stable:

$$\Gamma_y(\omega) = |W(e^{j\omega})|^2 \Gamma_v(\omega)$$

Proof. It will not be done

□

Note it has the same hypothesis as theorem 4.9

Remark.

■

$W(z)|_{z=e^{j\omega}} = W(e^{j\omega})$ is called “frequency response” of $W(z)$.

Example.

MA(1): $y(t) = e(t) + c \cdot e(t-1)$, $c \in \mathbb{R}, e(t) \sim WN(0, 1)$.

- $\gamma_y(0) = (1^2 + c^2) \cdot 1 = 1 + c^2$
- $\gamma_y(1) = 1 \cdot c = c = \gamma_y(-1)$
- $\gamma_y(\tau) = 0 \quad \forall |\tau| > 1$

So:

$$\Gamma_y(\omega) = \sum_{\tau=-\infty}^{\infty} \gamma_y(\tau) e^{-j\omega\tau} = (1 + c^2) + c \cdot e^{-j\omega} + c \cdot e^{j\omega} = 1 + c^2 + c(e^{j\omega} + e^{-j\omega})$$

Remember that $e^{j\omega}$ is a vector turning in the complex plane:

$$\begin{aligned} e^{j\omega} &= \cos \omega + j \sin \omega \\ e^{-j\omega} &= \cos \omega - j \sin \omega \end{aligned}$$

Then:

$$\Gamma_y(\omega) = 1 + c^2 + 2c \cos \omega$$

Note that this is a real, even, positive, and 2π -periodic. So it satisfies the properties of these functions.

Now, use the spectral factorization theorem to find another way to get $\Gamma_y(\omega)$:

$$y(t) = e(t) + c \cdot e(t-1) = (1 + c \cdot z^{-1})e(t) = W(z)e(t)$$

Now, by the theorem 5.3:

$$\Gamma_y(\omega) = |W(e^{j\omega})|^2 \cdot \Gamma_e(\omega)$$

Note that $\Gamma_e(\omega) = \lambda^2 = 1$, and:

$$\begin{aligned} |W(e^{j\omega})|^2 &= |1 + ce^{-j\omega}|^2 = (1 + ce^{j\omega})(1 + ce^{-j\omega}) \\ &= 1 + c^2 \cdot e^{-j\omega} e^{j\omega} + ce^{-j\omega} + ce^{j\omega} \\ &= 1 + c^2 + c(e^{j\omega} + e^{-j\omega}) = 1 + c^2 + 2c \cdot \cos \omega \end{aligned}$$

We obtained the same result, so the theorem worked to calculate the $\Gamma_y(\omega)$.

We can graph this function taking as reference $\omega = 0, \pi/2$, and π . We have to separate the cases where $c > 0$ or $c < 0$.

Finally, calculate $\gamma_y(0)$:

$$\begin{aligned} \gamma_y(0) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma_y(\omega) e^{-j\omega \cdot 0} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + c^2 + 2c \cos \omega) d\omega \\ &= \frac{1}{2\pi} ((1 + c^2)\omega)_{-\pi}^{\pi} + 2c[\sin \omega]_{-\pi}^{\pi} \\ &= \frac{1}{2\pi} ((1 + c^2)\pi - (1 + c^2)(-\pi) + 2c(0 - 0)) \\ &= \frac{1}{2\pi} (2\pi(1 + c^2)) = 1 + c^2 \end{aligned}$$

Note that we have seen that an ARMA(1,1) process has different representations:

- Time domain:

$$y(t) = a_1 y(t-1) + \dots + a_m y(t-m) + c_0 e(t) + \dots + c_n y(t-n)$$

- Operatorial:

$$y(t) = \frac{C(z)}{A(z)} e(t)$$

- Probabilistic:

$$m_y, \gamma_y(\tau)$$

- Frequency-domain:

$$m_y, \Gamma_y(\omega)$$

6 Prediction

6.1 Different Representations

It is possible to represent a SP in several ways, but these representations are not unique.

Specifically, for any SSP $y(t)$ there may exist infinitely many $W(z)$ and $e(t)$ that produce $y(t)$ as an output

Example.

$$\begin{aligned} y(t) &= W(z)e(t) & e(t) &\sim WN(0, \lambda^2) \\ &= W(z)\frac{\alpha}{\alpha}e(t) = \tilde{W}(z)\tilde{e}(t) & \tilde{W}(z) &= \frac{1}{\alpha}W(z) \quad \wedge \quad \tilde{e}(t) = \alpha e(t) \sim WN(0, \alpha^2 \lambda^2) \end{aligned}$$

Example.

$$\begin{aligned} y(t) &= W(z)e(t) & e(t) &\sim WN(0, \lambda^2) \\ &= W(z)\frac{z^n}{z^n}e(t) = \tilde{W}(z)\tilde{e}(t) & \tilde{W}(z) &= z^n W(z) \quad \wedge \quad \tilde{e}(t) = z^{-n}e(t) = e(t-n) \sim WN(0, \alpha^2) \end{aligned}$$

Example.

If $p \in \mathbb{C}, |p| < 1$

$$\begin{aligned} y(t) &= W(z)e(t) & e(t) &\sim WN(0, \lambda^2) \\ &= W(z)\frac{z-p}{z-p}e(t) = \tilde{W}(z)\tilde{e}(t) & \tilde{W}(z) &= \frac{z-p}{z-p}W(z) \quad \wedge \quad \tilde{e}(t) = e(t) \sim WN(0, \alpha^2) \end{aligned}$$

Example.

Let be $(z - q)$ a monomial

$$\begin{aligned} y(t) &= W(z)e(t) \quad e(t) \sim WN(0, \lambda^2) \\ &= W(z)(z - q)e(t) = \tilde{W}(z)\tilde{e}(t) \quad \tilde{W}(z) = (z - \frac{1}{q})W(z) \quad \wedge \quad \tilde{e}(t) = \frac{z - q}{z - \frac{1}{q}}e(t) = G(z)e(t) \end{aligned}$$

Note that $\tilde{e}(t)$ is still a WN. Indeed, by the theorem 5.3, for any w:

$$\begin{aligned} \Gamma_{\tilde{e}}(w) &= \left| \frac{e^{jw} - q}{e^{jw} - \frac{1}{q}} \right|^2 \Gamma_e(w) \\ &= \left(\frac{e^{jw} - q}{e^{jw} - \frac{1}{q}} \right) \left(\frac{e^{-jw} - q}{e^{-jw} - \frac{1}{q}} \right) \lambda^2 \quad (\Gamma(w) = \lambda^2 \quad \forall w, \text{ since } e \text{ is WN}) \\ &= \frac{1 + q^2 - q(e^{jw} + e^{-jw})}{1 + \frac{1}{q^2} - \frac{1}{q}(e^{jw} + e^{-jw})} \lambda^2 = \frac{1 + q^2 - q(2 \cos w)}{1 + \frac{1}{q^2} - \frac{1}{q}(2 \cos w)} \lambda^2 \\ &= q^2 \lambda^2 \quad (\text{Factorizing the numerator by } q^2) \end{aligned}$$

Plus, we can derive another case where T is a system called all-pass filter:

$$T(z) = \frac{1}{q} \frac{z - q}{z - \frac{1}{q}} = \frac{1}{q} G(z)$$

Note that:

$$\Gamma_y(w) = \underbrace{|T(e^{jw})|^2}_1 \Gamma_e(w) = \Gamma_e(w)$$

So y and e are equivalent, but not equal.

6.2 Canonical Representation

In the following, we will need to define any SSP in a unique way. We will select the so-called canonical representation.

Theorem 6.1

Let $y(t)$ be an SSP with a rational $\Gamma_y(w)$. There exists a unique pair $(W(z), e(t))$ for y if and only if given $W(z) = \frac{C(z)}{A(z)}$, the following conditions hold:

- $C(z)$ and $A(z)$ are monic. i.e., can be written as follows:

$$C(z) = 1 + c_1 z^{-1} + \dots + c_n z^{-n}$$

$$A(z) = 1 + a_1 z^{-1} + \dots + a_m z^{-m}$$

- $C(z)$ and $A(z)$ have null relative degree in the polynomial representation:

$$\nu = m - n = 0$$

- $C(z)$ and $A(z)$ are coprime, i.e. they have no common factors.
- $C(z)$ and $A(z)$ have roots inside the unit circle (< 1).

Note that each condition could be achieved using the transformations done in the examples of section 6.1.

Definition 6.2: Canonical Representation

An SSP is said to be in the canonical form if there exists a unique pair $(W(z), e(t))$ for y .

6.3 Predictors

Given an SSP $y(t)$ in canonical form and a specific realization $y(0), y(1), \dots, y(t)$, we want to predict the future values of the process at a generic time $t+k$.

Let be

$$y(t) = \frac{C(z)}{A(z)} e(t) \quad e(t) \sim WN(\mu, \lambda^2)$$

with $C(z), A(z), \mu, \lambda^2$ known. Examples of predictors:

-

$$\hat{y}(t+k|t) = m_y = \frac{C(1)}{A(1)} \mu$$

-

$$\hat{y}(t+k|t) = y(t)$$

-

$$\hat{y}(t+k|t) = \frac{a_1 y(t) + a_2 y(t-1) + a_3 y(t-2)}{3}$$

- And others

Note that the first one uses only model information, while the second only uses the data. The third one uses both data and model information, but not every information.

6.4 Optimality Criterion

We would like $\hat{y}(t+k|t)$ to be optimal in some sense. We need to define an optimality criterion, exploiting all the information we have, both model and past data.

Definition 6.3: Prediction error

$$\varepsilon(t+k|t) := y(t+k) - \hat{y}(t+k|t)$$

Definition 6.4: Mean Squared Prediction Error

$$J(\hat{y}) = E[\varepsilon(t+k|t)^2] = E[(y(t+k) - \hat{y}(t+k|t))^2]$$

We need to find the predictor $\hat{y}(t+k|t)$ that minimizes $J(\hat{y})$. This is a hard problem to solve in the general case.

6.5 Reformulation in Linear Models

Anyway, we work with linear models (ARMA), and we can find a better problem representation.

Take a general predictor:

$$\hat{y}(t+k|t) = \alpha_0 y(t) + \alpha_1 y(t-1) + \dots = \sum_{i=0}^{\infty} \alpha_i y(t-i), \quad \sum_{i=0}^{\infty} \alpha_i < \infty$$

$$\hat{y}(t+k|t) = y(t) \sum_{i=0}^{\infty} \alpha_i z^{-i} = F_{\alpha}(z)y(t)$$

Note that the actual process is equivalent to a MA(∞) process:

$$\begin{aligned} y(t) &= W(z)e(t) & e(t) &\sim WN(0, \lambda^2) \\ &= \sum_{i=0}^{\infty} w_i e(t-i) \\ y(t-1) &= \sum_{i=0}^{\infty} w_i e(t-1-i) \\ &\dots \\ y(t-j) &= \sum_{i=0}^{\infty} w_i e(t-j-i) \end{aligned}$$

Therefore the predictor could be rewritten as:

$$\begin{aligned}
\hat{y}(t+k|t) &= \sum_{i=0}^{\infty} \alpha_i y(t-i) \\
&= \sum_{i=0}^{\infty} \alpha_i \sum_{j=0}^{\infty} w_j e(t-i-j) \\
&= \sum_{i=0}^{\infty} \beta_i e(t-i), \quad \text{where } \beta_i = \alpha_i \sum_{j=0}^{\infty} w_j z^{-j}
\end{aligned}$$

On the other side, note that the real element of the sequence can be written:

$$\begin{aligned}
y(t+k) &= \sum_{i=0}^{\infty} w_i e(t+k-i) \\
&= \underbrace{\sum_{i=0}^{k-1} w_i e(t+k-i)}_{(A)} + \underbrace{\sum_{i=0}^{\infty} w_{i+k} e(t-i)}_{(B)}
\end{aligned}$$

Note that (A) is the future, and (B) the past and present of $e(t)$

Then, the optimization problem can be represented equivalently as:

$$\hat{\beta} = \arg \min_{\beta} V(\beta) \quad \text{where } V(\beta) = E[(y(t+k) - \hat{y}(t+k|t, \beta))^2]$$

$$\begin{aligned}
V(\beta) &= E[(y(t+k) - \hat{y}(t+k|t, \beta))^2] \\
&= E \left[\left(\sum_{j=0}^{k-1} w_j e(t+k-j) + \sum_{i=0}^{\infty} w_{k+i} e(t-i) - \sum_{i=0}^{\infty} \beta_i e(t-i) \right)^2 \right] \\
&= E \left[\underbrace{\left(\sum_{j=0}^{k-1} w_j e(t+k-j) \right)^2}_{\text{We cannot change this}} + \left(\sum_{i=0}^{\infty} (w_{k+i} - \beta_i) e(t-i) \right)^2 \right]
\end{aligned}$$

Since $2 \left(\sum_{j=0}^{k-1} w_j e(t+k-j) \right) \left(\sum_{i=0}^{\infty} (w_{k+i} - \beta_i) e(t-i) \right) = 0$ (e(t)'s are uncorrelated at different time)

6.6 Optimum for Linear Models

Therefore, the optimization problem can be represented equivalently as:

$$\hat{\beta} = \arg \min_{\beta} E \left[\left(\sum_{i=0}^{\infty} (w_{k+i} - \beta_i) e(t-i) \right)^2 \right]$$

We can easily note that this is ≥ 0 , so the minimum β_i 's are

$$\hat{\beta}_i = w_{k+i} \quad i = 0, 1, \dots$$

So, the optimal linear predictor from the noise is:

$$\hat{y}(t+k|t) = \sum_{i=0}^{\infty} w_{k+i} e(t-i)$$

Note that this is a $MA(\infty)$ process.

6.7 Analysis of Optimal Predictors

Limitation of the Predictor:

- It depends on the past samples of $e(t)$
- It is described by infinite w_i
- $e(t)$ must be a White Noise

Now, see the operatorial representation of the process:

$$W(z) = \frac{C(z)}{A(z)} = \frac{1 + c_1 z^{-1} + c_2 z^{-2} + \dots + c_n z^{-n}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_m z^{-m}} = w_0 + w_1 z^{-1} + w_2 z^{-2} + \dots$$

And by k steps of long division by $A(z)$, we obtain:

$$\frac{C(z)}{A(z)} = E(z) + \frac{z^{-k} F(z)}{A(z)}$$

Where $E(z)$ is the exact polynomial of order k , and $z^{-k} F(z)$ is the rest.

Indeed, from here we can obtain the Diophantine equation:

$$C(z) = E(z)A(z) + z^{-k} F(z)$$

Now, we can represent the element to predict as:

$$\begin{aligned} y(t+k) &= \frac{C(z)}{A(z)} e(t+k) = \left[E(z) + z^{-k} \frac{F(z)}{A(z)} \right] \cdot e(t+k) \\ &= E(z) e(t+k) + \frac{F(z)}{A(z)} z^{-k} e(t+k) \end{aligned}$$

The first term is a polynomial of order k : $e_0 + e_1 z^{-1} + \dots + e_k z^{-k}$. This times $e(t+k)$ represents future samples of $e(t)$, so it is unpredictable at time t . This is called the prediction error.

$e_0 = 1$ if C and A are in the canonical form.

The second term depends on $e(t)$, since $z^{-k} e(t+k) = e(t)$. This is the term we can minimize. This is the prediction of y at time t . It can be shown that the predictor can be written as:

$$\hat{y}(t+k|t) = \frac{F(z)}{A(z)} e(t)$$

6.8 Reconstruct $e(t)$

The problem is that we still depend on $e(t)$, and the observable measures are over $y(t)$. We need to reconstruct $e(t), e(t-1), \dots$ based on $y(t), y(t-1), \dots$.

Note that $y(t) = \frac{C(z)}{A(z)}e(t)$. Then,

$$e(t) = \frac{A(z)}{C(z)}y(t)$$

is called the whitening filter.

Note that:

$$\hat{y}(t+k|t) = \frac{F(z)}{A(z)}e(t) = \frac{F(z)}{A(z)}\frac{A(z)}{C(z)}y(t) = \frac{F(z)}{C(z)}y(t)$$

So the predictor depends on the data y , and not e .

Remark.

- $\hat{y}(t+k|t, s) = \frac{F(z)}{C(z)}y(t)$ is a stochastic process.
- $\hat{y}(t+k|t, s)$ is stationary because $C(z)$ has roots inside the unit circle in the canonical form. This implies that:

$$\hat{y}(t+k|t, s) = \frac{F(z)}{C(z)}y(t) \iff \hat{y}(t|t-k, s) = \frac{F(z)}{C(z)}y(t-k)$$

- The quality of the prediction gets worse as increasing k .

Note that the prediction error is a MA(k-1):

$$\begin{aligned} \varepsilon(t+k|t) &= y(t+k) - \hat{y}(t+k|t) = E(z)e(t+k) \\ &= e_0e(t) + e_1e(t-1) + \dots + e_{k-1}e(t-k+1) \end{aligned}$$

Then:

$$E[\varepsilon(t+k|t)^2] = (e_0^2 + e_1^2 + \dots + e_{k-1}^2)\lambda^2$$

It can be seen that this increases as k increases. Anyway, this has an asymptotic value:

$$Var(MA(\infty)) = Var(y(t)) \quad (\text{Since } y(t)=MA(\infty), \text{ see section 6.5})$$

6.9 Comparison with Trivial Predictor

Trivial predictor:

$$\begin{aligned} \hat{y}(t+k|t) &= E[y(t)] = m_y \\ \varepsilon(t+k|t)_{trivial} &= y(t+k) - \hat{y}(t+k|t) = y(t+k) - m_y \\ E[\varepsilon(t+k|t)_{trivial}^2] &= E[(y(t+k) - \hat{y}(t+k|t))^2] = Var[y(t)] \end{aligned}$$

Note that the error of our predictor satisfies:

$$\text{Var}[e(t)] \leq \text{Var}[\varepsilon(t+k|t)] < \text{Var}[y(t)] = \text{Var}[\varepsilon(t+k|t)_{\text{trivial}}]$$

So our predictor is better than the trivial predictor

6.10 Starting point

When we start to predict elements, we don't have past predictions or data that we can rely on. Therefore, the common solution is to set:

$$\hat{y}(0|-1) = E[y(t)] = m_y$$

Anyway, any value is acceptable, since $\frac{F(z)}{C(z)}$ is asymptotically stable, and any initial condition vanishes rapidly.

6.11 Non-zero mean case

Now, we would like to manage the case of non-zero mean WN as well:

$$y(t) = \frac{C(z)}{A(z)}e(t) \quad e(t) \sim WN(\mu, \lambda^2)$$

Note that the mean of y is:

$$E[y(t)] = m_y = W(1)\mu = \frac{C(1)}{A(1)}\mu$$

Now, the predictor for this case is different. We consider the unbiased processes:

$$\begin{cases} \tilde{y}(t) = y(t) - m_y \\ \tilde{e}(t) = e(t) - \mu \end{cases}$$

Remember that $\tilde{y}(t) = W(z)\tilde{e}(t)$, $m_{\tilde{y}} = 0$.

Using the zero-mean case:

$$\hat{\tilde{y}}(t+k|t) = \frac{F(z)}{C(z)}\tilde{y}(t)$$

Note that m_y is constant, the process $\tilde{y}$ is y shifted by this value, so:

$$\hat{y}(t+k|t) = \hat{\tilde{y}}(t+k|t) + m_y$$

Now:

$$\begin{aligned} \hat{y}(t+k|t) &= \frac{F(z)}{C(z)}\tilde{y}(t) + m_y = \frac{F(z)}{C(z)}(y(t) - m_y) + m_y \\ &= \frac{F(z)}{C(z)}y(t) - \frac{F(z)}{C(z)}m_y + m_y \\ &= \frac{F(z)}{C(z)}y(t) + \text{bias} \quad \text{bias} = \left(1 - \frac{F(z)}{C(z)}\right)m_y \end{aligned}$$

We can consider the bias as constant since m_y is constant. Plus, the steady-state response of $\frac{F(z)}{C(z)}m_y$ is constant: $\frac{F(1)}{C(1)}m_y$. This is called the “static gain” of the system

6.12 ARMAX process case

An ARMAX process is characterized by:

$$y(t) = \frac{C(z)}{A(z)}e(t) + \frac{B(z)}{A(z)}u(t-d)$$

Note that the second term is fully deterministic. $u(t)$ is a real measured signal, and $\frac{B(z)}{A(z)}$ is the model of a physical system.

Plus, we have to assume that:

- $C(z)$ and $A(z)$ are canonical
- $u(t)$ is known from $t \rightarrow -\infty$ until $t \rightarrow \infty$

Now, define a new process:

$$z(t) = y(t) - \frac{B(z)}{A(z)}u(t-d) = \frac{C(z)}{A(z)}e(t)$$

By the previous case, the predictor of $z(t+k)$ is:

$$\hat{z}(t+k|t) = \frac{F(z)}{C(z)}z(t)$$

And now, as the rest is deterministic, we can take the predictor of $y(t+k)$ as:

$$\hat{y}(t+k|t) = \hat{z}(t+k|t) + \frac{B(z)}{A(z)}u(t-d)$$

Now, we would like to write $\hat{z}(t+k|t)$ as a function of the model and data $(u(t), y(t))$:

$$\begin{aligned} \hat{y}(t+k|t) &= \frac{B(z)}{A(z)}u(t+k-d) + \hat{z}(t+k|t) = \frac{B(z)}{A(z)}u(t+k-d) + \frac{F(z)}{C(z)}z(t) \\ &= \frac{B(z)}{A(z)}u(t+k-d) + \frac{F(z)}{C(z)}\left(y(t) - \frac{B(z)}{A(z)}u(t-d)\right) \\ &= \frac{F(z)}{C(z)}y(t) + \frac{B(z)}{C(z)}\left[\frac{C(z)}{A(z)} - \frac{F(z)}{A(z)}z^{-k}\right]u(t+k-d) \\ &= \frac{F(z)}{C(z)}y(t) + \frac{B(z)E(z)}{C(z)}u(t+k-d) \quad (\text{By the diophantine equation}) \end{aligned}$$

Then, the final predictor actually depends on:

- The past data of $u(t)$ and $y(t)$
- Model Information $(C(z), B(z), E(z), F(z))$

7 Identification Methodology

7.1 Problem

The identification problem consists of describing a system S that has inputs $u(t)$ and outputs $y(t)$. This description is done using the information of N examples $(u(1), y(1)), (u(2), y(2)), \dots, (u(N), y(N))$. In particular, we will focus on parametric system identification. This is done by steps:

1. Experiment Design (and Data Collection)
2. Selection of a Parametric Model Class $M(\theta) = \{M_\theta \text{ st } \theta \in \Theta\}$
3. Choice of the identification criterion $J_N(\theta) \geq 0$
4. Minimization of $J_N(\theta)$ wrt θ and finding this argmin $\hat{\theta}_N$:

$$\hat{\theta} = \arg \min_{\theta} J_N(\theta)$$

5. Model Validation

7.2 Experiment Design

1. The first step of the experiment design is designing a $u(t)$, the input of S . This gives informativity of the resulting dataset.
2. Select the experiment length N . This gives confidence in the result.

7.3 Choice of the Model Class

$$M(\theta) = \{M_\theta \text{ st } \theta \in \Theta\}$$

We can choose between diverse options of model classes:

- Discrete-time or continuous-time
- Linear or nonlinear
- Time-invariant or time-varying
- Static or dynamic

We will be focused on models with discrete-time, linear, time-invariant, and dynamic. In particular, ARMA and ARMAX models are within this class:

$$y(t) = a_1 y(t-1) + a_2 y(t-2) + \dots + b_0 y(t-d) + \dots + c_0 e(t) +$$

$$\theta = \begin{bmatrix} a_1 & a_2 & \dots & a_m & b_0 & \dots & b_p & c_0 & \dots & c_n \end{bmatrix}^T$$

Anyway, to fully specify $y(t)$ as an ARMAX process θ is not sufficient. We need also:

- λ^2 , coming from $e(t) \sim WN(0, \lambda^2)$
- d : Pure delay between $u(t)$ and $y(t)$

- Degrees of the polynomial $A(z), B(z), C(z) \rightarrow m, n, p$.

This is the most critical one, m,n, and p need to be specified a “priori”

Plus, note that Θ is the set of admissible values of θ .

7.4 Choice of $J_N(\theta) \geq 0$

By the predictive approach:

$$J(\theta) = E[y(t+1) - \hat{y}(t+1|\theta)]^2$$

but this is not computable, since we have only one realization.

We are going to use a sample-based version of the ideal object:

$$J_N(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1, \theta))^2$$

If the model is exact and we compute the optimal 1-step ahead prediction error:

$$\varepsilon(t+1|t) = e(t) \quad \text{and} \quad \text{Var}[\varepsilon(t+1|t)] = \text{Var}[e(t)] = \lambda^2$$

Then:

$$J(\hat{\theta}_{opt}) = E[e(t)] = \lambda^2$$

7.5 Optimization Problem

We need to minimize $J_N(\theta)$ wrt θ such that $J_N(\theta) : \Theta \subseteq \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}^+$.

There are two different situations:

- $J_N(\theta)$ is quadratic (AR, ARX)
- $J_N(\theta)$ is non-quadratic (ARMA, ARMAX)

7.6 Model Validation

Firstly, we made a few assumptions:

- S lies within the model class M_θ .
- m,n, and p are specified a priori.
- Data have been “correctly” selected, they are informative.

Then, based on this, we need to evaluate the predictor created, i.e., check the quality of the model created.

8 Identification of AR and ARX models

8.1 Model Class

The model class in this case is:

$$M_\theta : y(t) = \frac{B(z)}{A(z)}u(t-d) + \frac{1}{A(z)}e(t), \quad e(t) \sim WN(0, \lambda^2)$$

where:

- $A(z) = 1 - a_1z^{-1} - a_2z^{-2} - \dots - a_mz^{-m}$
- $B(z) = b_0 + b_1z^{-1} + \dots + b_{p-1}z^{-p+1}$

Note that compared with a general ARMAX, here $C(z) = 1$. Plus, we are assuming $A(z)$ is in a canonical form.

8.2 Support

Note that θ has a size of $n_\theta = m + p$:

$$\theta = \begin{bmatrix} a_1 & a_2 & \dots & a_m & b_0 & \dots & b_{p-1} \end{bmatrix}^T$$

8.3 Data Collection

Finally, the dataset will be in the form:

$$\mathcal{D}_N = \{(u(1), y(1)), (u(2), y(2)), \dots, (u(N), y(N))\}$$

8.4 Rewriting model Class

On the other hand, we can rewrite the model class as:

$$\begin{aligned} M_\theta : A(z)y(t) &= B(z)u(t-d) + e(t) \\ A(z)y(t) &= B(z)u(t-d) + e(t) + y(t) - y(t) \end{aligned}$$

So:

$$\begin{aligned} M_\theta : y(t) &= (1 - A(z))y(t) + B(z)u(t-d) + e(t) \\ &= (a_1z^{-1} + a_2z^{-2} + \dots + a_mz^{-m})y(t) + (b_0 + b_1z^{-1} + \dots + b_{p-1}z^{-p+1})u(t-d) + e(t) \end{aligned}$$

Note that u and y are inside the dataset, but $e(t)$ is not available at time t .

8.5 Predictor

It can be shown that, following the general case shown in section 6.8, the predictor is:

$$\begin{aligned} \hat{y}(t|t-1) &= (a_1z^{-1} + a_2z^{-2} + \dots + a_mz^{-m})y(t) + (b_0 + b_1z^{-1} + \dots + b_{p-1}z^{-p+1})u(t-d) \\ &= a_1y(t-1) + \dots + a_my(t-m) + b_0u(t-d) + b_1u(t-d-1) + \dots + b_{p-1}u(t-d-p+1) \end{aligned}$$

8.6 Rewriting Predictor

We will rewrite this so the calculations will be easier. We need to introduce the regression vector:

$$\varphi(t) = \begin{bmatrix} y(t-1) & y(t-2) & \dots & y(t-m) & u(t-d) & \dots & u(t-d-p+1) \end{bmatrix}^T$$

So:

$$\hat{y}(t|t-1) = \begin{bmatrix} a_1 & a_2 & \dots & a_m & b_0 & \dots & b_{p-1} \end{bmatrix} \cdot \begin{bmatrix} y(t-1) \\ y(t-2) \\ \dots \\ y(t-m) \\ u(t-d) \\ \dots \\ u(t-d-p+1) \end{bmatrix} = \theta^T \varphi(t)$$

8.7 Optimization task

Now, the optimization task is:

$$J_N(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1, \theta))^2$$

Replacing the rewritten predictor, J_N changes to:

$$\begin{aligned} J_N(\theta) &= \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1, \theta))^2 \\ &= \frac{1}{N} \sum_{t=1}^N (y(t) - \varphi(t)^T \theta)^2 \end{aligned}$$

Note that $\hat{y}$ is linear in θ , so J_N is quadratic in θ . This means that the function is convex, and therefore there's a unique minimum, if exists.

8.8 Solution Existence

Anyway, the solution exists if the following conditions are satisfied:

- $\hat{\theta}_N$ is a stationary point of $J_N(\theta)$:

$$\left. \frac{\partial J_N(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}_N} = 0$$

- $\hat{\theta}_N$ is a minimum point:

$$\left. \frac{\partial^2 J_N(\theta)}{\partial \theta^2} \right|_{\theta=\hat{\theta}_N} \succ 0$$

In particular, for our case:

$$\begin{aligned}
\frac{\partial J_N(\theta)}{\partial \theta} &= \frac{d}{d\theta} [J_N(\theta)] = \frac{d}{d\theta} \left[\frac{1}{N} \sum_{t=1}^N (y(t) - \varphi(t)^T \theta)^2 \right] \\
&= \frac{1}{N} \sum_{t=1}^N \frac{d}{d\theta} [(y(t) - \varphi(t)^T \theta)^2] \\
&= \frac{1}{N} \sum_{t=1}^N \left[2(y(t) - \varphi(t)^T \theta) \cdot \frac{d}{d\theta} (y(t) - \varphi(t)^T \theta) \right] \\
&= \frac{1}{N} \sum_{t=1}^N [2(y(t) - \varphi(t)^T \theta) \cdot -\varphi(t)]
\end{aligned}$$

So, to satisfy the first condition:

$$\begin{aligned}
0 &= \left. \frac{\partial J_N(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}_N} \\
&= \frac{1}{N} \sum_{t=1}^N [2(y(t) - \varphi(t)^T \hat{\theta}_N) \cdot -\varphi(t)]
\end{aligned}$$

So

$$\sum_{t=1}^N \varphi(t) y(t) = \left[\sum_{t=1}^N \varphi(t) \varphi(t)^T \right] \hat{\theta}_N$$

If $\sum_{t=1}^N \varphi(t) \varphi(t)^T$ (called information matrix) is non-singular:

$$\hat{\theta}_N = \left[\sum_{t=1}^N \varphi(t) \varphi(t)^T \right]^{-1} \left(\sum_{t=1}^N \varphi(t) y(t) \right)$$

This is the ordinary least squares formula, an explicit solution for the linear identification problem, but only if it is a minimum, and to verify this, we have to find:

$$\begin{aligned}
\frac{\partial J_N(\theta)}{\partial \theta} &= \frac{1}{N} \sum_{t=1}^N [2(y(t) - \varphi(t)^T \theta) \cdot -\varphi(t)] = -\frac{2}{N} \sum_{t=1}^N \varphi(t) y(t) + \frac{2}{N} \sum_{t=1}^N \varphi(t) \varphi(t)^T \theta \\
\frac{\partial^2 J_N(\theta)}{\partial \theta^2} &= \frac{d}{d\theta} \left[\frac{\partial J_N(\theta)}{\partial \theta} \right] = \frac{2}{N} \sum_{t=1}^N \varphi(t) \varphi(t)^T
\end{aligned}$$

And must be satisfied $\frac{\partial^2 J_N(\theta)}{\partial \theta^2} \succ 0$.

Therefore, we need to check that the information matrix is positive definite:

$$\begin{aligned}
x^T \sum_{t=1}^N \varphi(t) \varphi(t)^T x &= \sum_{t=1}^N x^T \varphi(t) \varphi(t)^T x \\
&= \sum_{t=1}^N (\varphi(t)^T x)^2 \geq 0
\end{aligned}$$

Therefore we can have two cases:

$$\frac{\partial^2 J_N(\theta)}{\partial \theta^2} \succ 0 \quad \text{and} \quad \frac{\partial^2 J_N(\theta)}{\partial \theta^2} = 0$$

The second case is called the degenerated case, and it does not assure finding a minimum point.

The reasons that can trigger a degenerate case are:

- Experimental Identifiability Issue: Data are not representative of the full underlying physical process.
- Structural Identifiability issue: The selected model class is too large, i.e., there is more than one model representing the same data. In this case, we must reduce the model class size.

9 Identification of ARMA and ARMAX models

9.1 Model Class

The model class in this case is:

$$M_\theta : y(t) = \frac{B(z)}{A(z)}u(t-d) + \frac{C(z)}{A(z)}e(t), \quad e(t) \sim WN(0, \lambda^2)$$

where:

- $C(z) = 1 + c_1 z^{-1} + c_2 z^{-2} + \dots + c_n z^{-n}$
- $A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - \dots - a_m z^{-m}$
- $B(z) = b_0 + b_1 z^{-1} + \dots + b_{p-1} z^{-p+1}$

9.2 Model Class

Note that θ has a size of $n_\theta = m + p + n$:

$$\theta = \begin{bmatrix} a_1 & a_2 & \dots & a_m & b_0 & \dots & b_{p-1} & c_1 & \dots & c_n \end{bmatrix}^T$$

9.3 Predictor

Now, remember that the predictor is:

$$\hat{y}(t|t-1, \theta) = \frac{B(z)E(z)}{C(z)}u(t-d) + \frac{F(z)}{C(z)}y(t-1)$$

Where $E(z)$ and $F(z)$ come from the k-step long division:

$$\frac{C(z)}{A(z)} = E(z) + \frac{z^{-k}F(z)}{A(z)}$$

Then, noting that the constant in $C(z)$ is 1, so the resulting polynomial $E(z)$ of order 1 and $F(z)$ for a one-step prediction are:

$$E(z) = 1 \quad \text{and} \quad F(z) = C(z) - A(z)$$

Substituting in the predictor:

$$\hat{y}(t|t-1, \theta) = \frac{C(z) - A(z)}{C(z)} y(t) + \frac{B(z)}{C(z)} u(t-d)$$

9.4 Predictor Error

Now, the predictor error is:

$$\begin{aligned} \varepsilon(t|t-1, \theta) &= y(t) - \hat{y}(t|t-1, \theta) = y(t) - \frac{C(z) - A(z)}{C(z)} y(t) - \frac{B(z)}{C(z)} u(t-d) \\ &= \left[1 - \frac{C(z) - A(z)}{C(z)} \right] y(t) - \frac{B(z)}{C(z)} u(t-d) \\ &= \frac{A(z)}{C(z)} y(t) - \frac{B(z)}{C(z)} u(t-d) \end{aligned}$$

θ appears in the denominator of ε , so we have a non-linear dependence, and we can't write the predictor as a linear expression. Therefore, this is a non-linear optimization problem.

9.5 Numerical Approach

We need an iterative numerical approach to solve this optimization problem, which has to have:

1. An initialization of the algorithm with an initial guess
2. An update rule $\theta^{(i+1)} = f(\theta^{(i)})$

such that the sequence converges to $\hat{\theta}_N = \arg \min_{\theta} J_N(\theta)$, and not to a local minimum.

9.5.1 Heuristic to avoid suboptimality

As iterative algorithms are guaranteed to converge to a local, and not to the global minimum, we will follow a heuristic approach that consists of:

- Using M initial guesses.
- Performing in each one N steps of the iterative algorithm.
- Between the

$$(\theta^{(N,1)}, \theta^{(N,2)}, \dots, \theta^{(N,M)})$$

each one with a

$$(J_N(\theta^{(N,1)}), J_N(\theta^{(N,2)}), \dots, J_N(\theta^{(N,M)}))$$

Choosing the θ with the lower cost

9.5.2 Update Rule

We will choose $\theta^{(i+1)}$ as the minimizer of:

$$V(\theta) = J_N(\theta^{(i)}) + \left. \frac{\partial J_N(\theta)}{\partial \theta} \right|_{\theta=\theta^{(i)}} (\theta - \theta^{(i)}) + \frac{1}{2} (\theta - \theta^{(i)})^T \left. \frac{\partial^2 J_N(\theta)}{\partial \theta^2} \right|_{\theta=\theta^{(i)}} (\theta - \theta^{(i)})$$

Note that this is a quadratic polynomial, so it is convex and we can just optimize it with the first-order conditions:

$$\frac{\partial V(\theta^{(i+1)})}{\partial \theta} = \left. \frac{\partial J_N(\theta)}{\partial \theta} \right|_{\theta=\theta^{(i)}} + \frac{1}{2} \left. \frac{\partial^2 J_N(\theta)}{\partial \theta^2} \right|_{\theta=\theta^{(i)}} (\theta^{(i+1)} - \theta^{(i)}) = 0$$

So we obtain the update rule, called Newton's law:

$$\theta^{(i+1)} = \theta^{(i)} - \underbrace{\left[\left. \frac{\partial^2 J_N(\theta)}{\partial \theta^2} \right|_{\theta=\theta^{(i)}} \right]^{-1}}_{\text{Inverse of the hessian on } \theta^{(i)}} \underbrace{\left. \frac{\partial J_N(\theta)}{\partial \theta} \right|_{\theta=\theta^{(i)}}}_{\text{Gradient in } \theta^{(i)}}$$

9.5.3 Computation of the derivatives

The first derivative:

$$\begin{aligned} \frac{\partial J_N(\theta)}{\partial \theta} &= \frac{d}{d\theta} \left[\frac{1}{N} \sum_{t=1}^N \varepsilon(t|t-1, \theta)^2 \right] = \frac{1}{N} \sum_{t=1}^N \frac{d}{d\theta} \varepsilon(t|t-1, \theta)^2 \\ &= \frac{2}{N} \sum_{t=1}^N \varepsilon(t|t-1, \theta) \cdot \frac{\partial \varepsilon(t|t-1, \theta)}{\partial \theta} \end{aligned}$$

The second derivative:

$$\begin{aligned} \frac{\partial^2 J_N(\theta)}{\partial \theta^2} &= \frac{d}{d\theta} \left[\frac{\partial J_N(\theta)}{\partial \theta} \right] \\ &= \frac{d}{d\theta} \left[\frac{2}{N} \sum_{t=1}^N \varepsilon(t|t-1, \theta) \cdot \frac{\partial \varepsilon(t|t-1, \theta)}{\partial \theta} \right] \\ &= \frac{2}{N} \sum_{i=1}^N \frac{\partial \varepsilon(t|t-1, \theta)}{\partial \theta} \left(\frac{\partial \varepsilon(t|t-1, \theta)}{\partial \theta} \right)^T + \frac{2}{N} \sum_{i=1}^N \varepsilon(t|t-1, \theta) \cdot \frac{\partial^2 \varepsilon(t|t-1, \theta)}{\partial \theta^2} \end{aligned}$$

Note that the second term is not convex, so its use can make the iteration unstable. We will just ignore it.

Now, we are missing $\frac{\partial \varepsilon(t|t-1, \theta)}{\partial \theta}$. We need to split the results on the different parameters of θ

- $a_i, \quad i = 1, \dots, m$

$$\begin{aligned} \frac{\partial \varepsilon(t|t-1, \theta)}{\partial a_i} &= \frac{\partial}{\partial a_i} \left[\frac{1 - a_1 z^{-1} - a_2 z^{-2} - \dots - a_m z^{-m}}{C(z)} y(t) - \frac{B(z)}{C(z)} u(t-d) \right] \\ &= \frac{\partial}{\partial a_i} \left[\frac{-a_i z^{-i}}{C(z)} y(t) \right] = -\frac{z^{-i}}{C(z)} y(t) =: \alpha(t-i) \end{aligned}$$

- $b_i, \quad i = 0, \dots, p-1$

$$\begin{aligned} \frac{\partial \varepsilon(t|t-1, \theta)}{\partial b_i} &= \frac{\partial}{\partial b_i} \left[\frac{A(z)}{C(z)} y(t) - \frac{b_0 + b_1 z^{-1} + \dots + b_{p-1} z^{-p+1}}{C(z)} u(t-d) \right] \\ &= \frac{\partial}{\partial b_i} \left[\frac{-b_i z^{-i}}{C(z)} u(t-d) \right] = -\frac{z^{-i}}{C(z)} u(t-d) =: \beta(t-i) \end{aligned}$$

- $c_i, \quad i = 1, \dots, n$

$$\frac{\partial \varepsilon(t|t-1, \theta)}{\partial c_i} = \frac{\partial}{\partial c_i} \left[\frac{A(z)}{1 + c_1 z^{-1} + c_2 z^{-2} + \dots + c_n z^{-n}} y(t) - \frac{B(z)}{1 + c_1 z^{-1} + c_2 z^{-2} + \dots + c_n z^{-n}} u(t-d) \right]$$

So:

$$\frac{\partial C(z) \varepsilon(t|t-1, \theta)}{\partial c_i} = A(z) y(t) - B(z) u(t-d) = 0$$

Now, using the product rule:

$$\frac{\partial C(z)}{\partial c_i} \varepsilon(t|t-1, \theta) + C(z) \frac{\partial \varepsilon(t|t-1, \theta)}{\partial c_i} = 0$$

Isolating the derivative:

$$\frac{\partial \varepsilon(t|t-1, \theta)}{\partial c_i} = -\frac{1}{C(z)} \frac{\partial C(z)}{\partial c_i} \cdot \varepsilon(t|t-1, \theta) = -\frac{1}{C(z)} z^{-i} \cdot \varepsilon(t|t-1, \theta) =: \gamma(t-i)$$

Finally,

$$\frac{\partial \varepsilon(t|t-1, \theta)}{\partial \theta} = \begin{bmatrix} \alpha(t-1) & \dots & \alpha(t-m) & \beta(t-d) & \dots & \beta(t-d-p+1) & \dots & \gamma(t-1) & \dots & \gamma(t-n) \end{bmatrix}$$

where:

$$\alpha(t) = -\frac{1}{C(z)} y(t) \quad \beta(t) = -\frac{1}{C(z)} u(t-d) \quad \gamma(t) = -\frac{1}{C(z)} \cdot \varepsilon(t|t-1, \theta)$$

So, replacing the derivatives on $\theta^{(i+1)}$, we have a closed expression. for the newton's rule.

9.5.4 Variations

There are some variations to the newton's rule:

- The Newton's Rule uses:

$$\theta^{(i+1)} = \theta^{(i)} - [hessian]^{-1} \cdot gradient$$

- The Gradient Descent uses a fixed and scalar step:

$$\theta^{(i+1)} = \theta^{(i)} - \eta \cdot gradient$$

- There also exists the Quasi-Newton's rule, that uses an approximated hessian, omitting

the inestable part of the second derivative of $J_N(\theta)$:

$$\theta^{(i+1)} = \theta^{(i)} - [app \ hessian]^{-1} \cdot gradient$$

We use the Quasi-Newton's rule. It is computationally lighter than newton's. Anyway it is more accurate and slower than Gradient Descent.

10 Model Validation

10.1 The Optimum and Convergence

Let be data $\mathcal{D}_N$, model class M_θ , predictor $\hat{\theta}_N$, error $\varepsilon(t|t-1, \theta)$, and optimization task $J_N(\theta)$. We want to test if $M(\hat{\theta}_N)$ is a “good” model for the system.

From now, we will assume that $N \rightarrow \infty$, so the information in every sub-dataset is the same.

Definition 10.1: global minimum points of $\bar{J}(\theta)$

$$\Delta = \{\theta^* \quad st \quad \bar{J}(\theta^*) \leq \bar{J}(\theta) \quad \forall \theta\}$$

Theorem 10.2

Under current Assumptions, as $N \rightarrow \infty$

$$J_N(\theta, S) \xrightarrow{N \rightarrow \infty} \bar{J}(\theta) = E_S[\varepsilon(t|t-1, \theta)^2]$$

Moreover, it is satisfied with probability 1:

$$\hat{\theta}_N(S) \xrightarrow{N \rightarrow \infty} \Delta$$

Proof. It won't be proved □

Corollary 10.3

If Δ is a singleton ($= \{\theta^*\}$), i.e., $\bar{J}(\theta)$ has a unique minimum point,

$$\implies \hat{\theta}_N(S) \xrightarrow{N \rightarrow \infty} \theta^*$$

Definition 10.4: θ^0

If exists, θ^0 is the element of the model class that represent to S:

$$S \equiv M(\theta^0)$$

Moreover, we are interested in guarantee that if $S \in M_\theta$, i.e. $\exists \theta^0 \quad st \quad S \equiv M(\theta^0)$, then $\hat{\theta}_N \rightarrow \theta^0$, i.e. our predictor converges to the element that represent S in M_θ

Theorem 10.5

If $S \in M_\theta$

$$\theta^0 \in \Delta$$

Proof.

$$\begin{aligned} \varepsilon(t|t-1, \theta) &= y(t) - \hat{y}(t|t-1, \theta) \\ &= \underbrace{y(t) - \hat{y}(t|t-1, \theta^0)}_{=e(t), \text{ since it is opt.}} + \hat{y}(t|t-1, \theta^0) - \hat{y}(t|t-1, \theta) \end{aligned}$$

Now, note that:

$$\begin{aligned} \bar{J}(\theta) &= \text{Var}[\varepsilon(t|t-1, \theta)] = E[\varepsilon(t|t-1, \theta)^2] \\ &= E[(e(t) + \hat{y}(t|t-1, \theta^0) - \hat{y}(t|t-1, \theta))^2] \\ &= E[e^2(t)] + E[(\hat{y}(t|t-1, \theta^0) - \hat{y}(t|t-1, \theta))^2] + 2E[e(t)(\hat{y}(t|t-1, \theta^0) - \hat{y}(t|t-1, \theta))] \\ &= E[e^2(t)] + E[(\hat{y}(t|t-1, \theta^0) - \hat{y}(t|t-1, \theta))^2] \quad (\text{Predictors are based until data } t-1, \text{ so it is uncorrelated}) \\ &= \underbrace{\lambda^2}_{=\bar{J}(\hat{\theta})} + \underbrace{E[(\hat{y}(t|t-1, \theta^0) - \hat{y}(t|t-1, \theta))^2]}_L \end{aligned}$$

Now, $L \geq 0$ and it is minimum for $\theta = \theta^0$. As λ^2 is deterministic:

$$\min_{\theta} \bar{J}(\theta) = \bar{J}(\theta^0) \leq \bar{J}(\theta) \quad \forall \theta \in \Theta$$

So θ^0 is minimum and then $\theta^0 \in \Delta$. □

Anyway, not always $S \in M_\theta$. There are 4 possible scenarios:

1. $S \in M_\theta$ and Δ is a singleton. ($S \equiv M(\theta^0), \Delta = \{\theta^*\}$).
In this scenario, $M(\hat{\theta}_N)$ converges to $M(\theta^0)$.
2. $S \in M_\theta$ and Δ is not a singleton.
In this scenario, $M(\hat{\theta}_N)$ converges to some point $\in \Delta$, but it is not necessarily $M(\theta^0)$.
3. $S \notin M_\theta$ and Δ is a singleton.
In this scenario, $M(\hat{\theta}_N)$ converges to some point $\in M_\theta$ such that the distance to $M(\theta^0)$ is minimum.
4. item $S \notin M_\theta$ and Δ is not a singleton.
In this scenario, $M(\hat{\theta}_N)$ converges to some point $\in \Delta$, but $M(\theta^0)$ is not inside, and it could be not the closest to $M(\theta^0)$.

10.2 Model Order Selection

$$M_\theta = \{M(\theta) \quad st \quad \theta \in \Theta \subseteq \mathbb{R}^{n_\theta}\}.$$

We cannot guarantee that $S \in M_\theta$, as S is unknown. Plus, we don't know how to enlarge M_θ if $S \notin M_\theta$. Anyway, we can find some heuristics.

For Simplicity, we will consider $n = m = p$ (orders of elements in the transfer functions).

Now, we want to select n properly such that $S \in M_\theta$.

We could simply take the following procedure:

Algorithm 1 Choose n

Require:

Ensure: n

$n=1$

for $n < n_{\max}$ **do**

$M_\theta^{(n)} = \{M(\theta), \theta \in \Theta^{(n)} \subseteq \mathbb{R}^{n_\theta}\}$

$J_N^{(n)}(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1, \theta^{(n)}))^2$

$\hat{\theta}_N^{(n)} = \arg \min_\theta J_N^{(n)}(\theta)$

$n+ = 1$

end for

But S cannot be detected just by looking at the behavior of $J_N(\hat{\theta}_N^{(n)})$ as a function of n . We can mistake on both sides, under and overfitting.

We can use 3 different criteria order for order model selection:

1. Whiteness test on residuals
2. Cross-Validation
3. Identification with model order penalties

10.3 Whiteness Test on Residuals

We know that if $S \in M_\theta$, exists n such that

$$\hat{\theta}_N^{(n)} \rightarrow \theta^0 \quad \wedge \quad \varepsilon(t|t-1, \hat{\theta}_N^{(n)}) \rightarrow e(t) \sim WN(0, \lambda^2)$$

Therefore, for a large N , $\varepsilon(t|t-1, \hat{\theta}_N^{(n)}) = y(t) - \hat{y}(t|t-1, \hat{\theta}_N^{(n)})$ should be a realization of a WN.

In theory:

- $\gamma_\varepsilon(\tau)$ should be λ^2 in 0 and 0 in other τ s.
- $\Gamma_\varepsilon(w)$ should be a constant function equal to λ^2 between $-\pi$ and π .

In practice, the behavior of the functions will be similar, but not exactly as they are supposed to be. What we consider close enough to the ideal functions is an arbitrary criterion.

Besides that, in the ideal situation, for a certain n , the ε will start to be a WN from the point where the real system order is. In the real world, this does not happen.

The Whiteness Test on Residuals, also known as Anderson's Test, consists of testing the above-mentioned conditions. Anyway, as we have seen, in the practice it does not give good results.

10.4 Cross-Validation

Given:

$$\mathcal{D}_N = \{(u(1), y(1)), \dots, (u(N), y(N))\}$$

Cross validation consists of choosing k elements to training and do identification, and using the rest as a validation test.

Algorithm 2 Cross-Validation

```

for  $n < n_{\max}$  do
   $M_{\theta}^{(n)} = \{M(\theta), \theta \in \Theta^{(n)} \subseteq \mathbb{R}^{n_{\theta}}\}$ 
   $J_K^{(n)}(\theta) = \frac{1}{K} \sum_{t=1}^K (y(t) - \hat{y}(t|t-1, \theta_K^{(n)}))^2$ 
   $\hat{\theta}_N^{(n)} = \arg \min_{\theta} J_K^{(n)}(\theta)$ 
  Evaluate  $J_V(\hat{\theta}_K^{(n)}) = \frac{1}{N-K} \sum_{t=K+1}^N (y(t) - \hat{y}(t|t-1, \hat{\theta}_K^{(n)}))^2$ 
   $n+ = 1$ 
end for
Choose  $n$  as the one minimizing  $J_v$ 

```

The function $J_V(\hat{\theta}_K^{(n)})$ is not not-increasing as J , so we can choose the minimum point, since there is a point where it starts to overfit and $J_V(\hat{\theta}_K^{(n)})$ starts to increase.

Note that we want both K and $(N - K)$ to be large, since the first one allows the model to be complex, and the second one allows the model estimation to be close to the real one.

In Machine Learning, k -fold cross-validation is often used. It can't be used in our type of problems, since the data is correlated by time.

10.5 Model Order Penalties

This strategy uses only one dataset, but using different costs for order model selection and parameter estimation.

Instead of minimizing $J_N(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1, \theta))^2$, we minimize another function, for example:

- **Final Prediction Error (FPE):**

The ideal objective is to find:

$$\min_{\theta} E[J_N(\theta)]$$

So FPE calculates a numerical approximation of this:

$$FPE(n) = \frac{N+n}{N-n} J_N(\hat{\theta}_N^{(n)})$$

- **Akaike Information Criterion (AIC):**

It is a criterion that comes from information theory.

$$AIC(n) = \ln(J_N(\hat{\theta}_N^{(n)})) + 2 \frac{n}{N}$$

The first term decreases with n . but the second term is a penalty for large n 's.

- **Minimum Description Length (MDL):**

It comes from AIC.

$$MDL(n) = \ln(J_N(\hat{\theta}_N^{(n)})) + \ln(N) \frac{n}{N}$$

Remark.

Note that:

$$\text{Remark: } FPE(n) = \ln \left(\frac{N+n}{N-n} J_N(\hat{\theta}_N^{(n)}) \right) = \ln \left(\frac{1 + \frac{n}{N}}{1 - \frac{n}{N}} J_N(\hat{\theta}_N^{(n)}) \right)$$

And also note that, comparing AIC and MDL:

If $\ln(N) > 2 \iff \ln \left(1 + \frac{n}{N} \right) + \ln \left(1 - \frac{n}{N} \right) + \ln \left(J_N(\hat{\theta}_N^{(n)}) \right)$ by MDL. From a practical point of view, this condition is always met (since e^2 is small, $\ln(1+x) \rightarrow x$)

It can be proved that if $S \in M_\theta$ and M_θ is the set of ARX models, MDL is right. In the general case, $S \notin M_\theta$. We usually prefer to slightly overfit, and use AIC.

11 Non-Parametric Identification

11.1 The problem

Starting from a portion of a single realization of a SSP, we may want to infer some statistical properties of it. e.g.: $m_u, \gamma_y(\tau), \Gamma_y(w)$.

We already know how to do this with enough data with a “model-based” inference:

$$\mathcal{D}_N = \{y(1), y(2), \dots, y(N)\}$$

We have to identify the model (find its transfer function $W(z)$) and calculate the statistics. The problem with this approach is that in order to get the statistical properties, we have to estimate something that is much more complicated, we may find an easier way to do it.

11.2 General View

Non-Parametric identification means that we don’t have to find the parameters of the model, but obtain the statistical properties directly.

Definition 11.1: Correctness

We say an estimator is correct if the expected value of the estimator is equal to the probabilistic property to be estimated. i.e., this definition measures the bias of the estimator

Definition 11.2: Consistency

We say an estimator is consistent if the variance of the estimator error tends to 0 as the number of data tends to infinite

11.3 Sample Mean

The direct estimator of m_y is:

$$\hat{m}_N = \frac{1}{N} \sum_{t=1}^N y(t)$$

This estimator $\hat{m}_N$ is correct if:

$$E_{\mathcal{D}_N}[\hat{m}_N] = m_y$$

It is actually correct:

$$E_{\mathcal{D}_N}[\hat{m}_N] = E \left[\frac{1}{N} \sum_{t=1}^N y(t) \right] = \frac{1}{N} \sum_{t=1}^N E[y(t)] = \frac{1}{N} \sum_{t=1}^N m_y = m_y$$

This estimator is consistent if:

$$E [(\hat{m}_N - m_y)^2] \xrightarrow{N \rightarrow \infty} 0$$

This is not always true, such as the following example:

Example.

Let be $y(t, s) = v(s) \quad \forall t, \quad (s) \sim N(0, 1)$.

$$\begin{aligned} E [(\hat{m}_N - m_y)^2] &= E \left[\left(\frac{1}{N} \sum_{t=1}^N y(t, s) - m_y \right)^2 \right] = E \left[\left(\frac{1}{N} \sum_{t=1}^N v(s) - 0 \right)^2 \right] \\ &= E \left[\left(\frac{1}{N} N v(s) \right)^2 \right] = E [v(s)^2] = 1 \quad \forall N \end{aligned}$$

Anyway, $\hat{m}_N$ is actually consistent in some cases:

Theorem 11.3

If $\gamma(\tau) \rightarrow 0$ as $|\tau| \rightarrow +\infty$:

$\hat{m}_N$ is consistent

Corollary 11.4

ARMA process $\hat{m}_N$'s are consistent.

11.4 Sample Covariance Function

Assume, without loss of generality, that $m_y = 0$.

The direct estimator of $\gamma_N(\tau)$ is:

$$\hat{\gamma}_N(\tau) = \frac{1}{N - |\tau|} \sum_{t=1}^{N-|\tau|} y(t)y(t + |\tau|), \quad 0 \leq |\tau| \leq N - 1$$

Note that:

$$\begin{aligned}\hat{\gamma}_N(0) &= \frac{1}{N} [y(1)^2 + \cdots + y(N)^2] \\ \hat{\gamma}_N(1) &= \frac{1}{N-1} [y(1)y(2) + \cdots + y(N-1)y(N)] \\ &\vdots \\ \hat{\gamma}_N(N-1) &= \frac{1}{1} [y(1)y(N)]\end{aligned}$$

Anyway, we cannot trust an estimator based only on ‘one sample’, so $\hat{\gamma}_N(N-1)$ is not reliable. Actually, we will say that the estimator will be reliable if $|\tau| \ll N-1$

Remark.

Note that τ is a even function, so $\hat{\gamma}_N(\tau) = \hat{\gamma}_N(-\tau)$

The estimator $\hat{\gamma}_N(\tau)$ is correct:

$$\begin{aligned}E[\hat{\gamma}_N(\tau)] &= E\left[\frac{1}{N-\tau} \sum_{t=1}^{N-\tau} y(t)y(t+\tau)\right] = \frac{1}{N-\tau} \sum_{t=1}^{N-\tau} E[y(t)y(t+\tau)] \\ &= \frac{1}{N-\tau} (N-\tau)\gamma(\tau) = \gamma(\tau)\end{aligned}$$

Plus, the estimator is consistent under some assumptions:

Theorem 11.5

If $\gamma(\tau) \rightarrow 0$ as $|\tau| \rightarrow +\infty$:

$\hat{\gamma}_N(\tau)$ is consistent

Corollary 11.6

ARMA process $\hat{\gamma}_N(\tau)$ ’s are consistent.

11.5 Sample Spectrum

One possible estimator is:

$$\hat{\Gamma}_y(w) = \sum_{t=-(N-1)}^{N-1} \hat{\gamma}_N(\tau) e^{-jw\tau}$$

Remark.

This is an indirect estimator since it is based on another estimator.

The two sources of approximations are the estimator $\gamma_N(\tau)$ and the fact that the sum is not on $[-\infty, \infty]$.

This estimator is not strictly correct:

$$\begin{aligned}
E \left[\hat{\Gamma}_y(w) \right] &= E \left[\sum_{t=-(N-1)}^{N-1} \hat{\gamma}_N(\tau) e^{-jw\tau} \right] = \sum_{t=-(N-1)}^{N-1} E \left[\hat{\gamma}_N(\tau) \right] e^{-jw\tau} \\
&= \sum_{t=-(N-1)}^{N-1} \gamma_N(\tau) e^{-jw\tau} \quad (\text{Assuming correctness of } \hat{\gamma}_N(\tau))
\end{aligned}$$

We can see, as the sum is on $[-N-1, N+1]$, this estimator is not correct, but asymptotically correct (under correctness of $\hat{\gamma}_N(\tau)$ assumption), i.e.:

$$\lim_{N \rightarrow \infty} E \left[\hat{\Gamma}_y(w) \right] = \Gamma_y(w) \quad \forall w$$

Despite this, in this case, the estimator is not consistent:

Theorem 11.7

The spectrum is not consistent, actually:

$$E \left[(\hat{\Gamma}_N(w) - \Gamma_y(w))^2 \right] \xrightarrow{N \rightarrow +\infty} \Gamma_y(w)^2 \quad \forall w$$

Moreover:

$$E \left[(\hat{\Gamma}_N(w_1) - \Gamma_y(w_1))(\hat{\Gamma}_N(w_2) - \Gamma_y(w_2)) \right] \xrightarrow{N \rightarrow +\infty} 0 \quad \forall w_1 \neq w_2$$

Then, some techniques can be used to improve this estimation:

Definition 11.8: Averaging (“Regularization”)

It consists of splitting the training data into d parts. Then, we can calculate a different estimator for each subdataset and merge them. We notate the i -th estimator as:

$$\hat{\Gamma}_{\frac{N}{d}}^{(i)}(w)$$

Then, the final estimator is:

$$\hat{\hat{\Gamma}}_N(w) = \frac{1}{d} \sum_{i=1}^d \hat{\Gamma}_{\frac{N}{d}}^{(i)}(w)$$

Theorem 11.9

If $\frac{N}{d}$ is large enough:

$$E \left[(\hat{\hat{\Gamma}}_N(w) - \Gamma_y(w))^2 \right] \simeq \frac{1}{d} E \left[(\hat{\Gamma}_N(w) - \Gamma_y(w))^2 \right]$$

So the $\hat{\hat{\Gamma}}$ is ‘more consistent’ than $\hat{\Gamma}$.

Remark.

If we have a d large, the estimate could be biased. Then, there exists a trade-off between consistency and correctness (Bias/variance trade-off).

On the other hand, the sample spectrum is typically computationally more demanding than the other sample estimators. To deal with this, we can take a different estimator of $\gamma_y(\tau)$:

Definition 11.10: $\hat{\gamma}'_N(\tau)$

$$\hat{\gamma}'_N(\tau) = \frac{1}{N} \sum_{t=1}^{N-\tau} y(t)y(t+\tau)$$

Theorem 11.11

$\hat{\Gamma}'_N(w)$ (The $\hat{\Gamma}'_N(w)$ calculated with $\hat{\gamma}'_N(\tau)$) is equivalent to:

$$\hat{\Gamma}'_N(\tau) = \frac{1}{N} \left| \sum_{t=1}^N y(t)e^{-jw\tau} \right|^2$$

Then, $\hat{\gamma}'_N(\tau)$ can be computed directly from $y(t)$. Plus, this is the Discrete Fourier Transform of $y(t)$, there are efficient ways to calculate it.

Note that this works, since for N large, $(N - \tau) \rightarrow N$

12 Data Preprocessing

12.1 Non-Stationary Time-Series

To work with non-stationary SP, we need first to estimate possible trends or seasonalities, then we can remove them from the original signal and work with the remaining SSP (See also section 7 of (Jeon, 2024)).

12.2 Trend Removal

Having a SP with a linear trend, we can decompose it:

$$y(t) = trend + SSP = k \cdot t + m + SSP$$

The only information available that we have is $\mathcal{D}_N$, so we can forecast this process as follows:

Algorithm 3 Forecast $y(t+1)$ with trend

Require: $\mathcal{D}_N$

Ensure: $\hat{y}(t+1|t)$

Estimate m and k : $(\hat{k}, \hat{m})$

$\tilde{y}(t) = y(t) - (\hat{k}t + \hat{m})$

$\hat{\tilde{y}}(t+1|t)$ from SSP formulas

$\hat{y}(t+1|t) = \hat{\tilde{y}}(t+1|t) + (\hat{k}(t+1) + \hat{m})$

Now, note that:

$$E[y(t) - kt - m] = E[\tilde{y}(t)] = 0$$

Then, we can estimate k and t with the Least Squares Ordinary formula:

$$(\hat{k}, \hat{m}) = \min_{k, m} \frac{1}{N} \sum_{t=1}^N (y(t) - kt - m)^2$$

Which the solution is:

$$\begin{bmatrix} \hat{k} \\ \hat{m} \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^N t^2 & \sum_{t=1}^N t \\ \sum_{t=1}^N t & N \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^N t \cdot y(t) \\ \sum_{t=1}^N y(t) \end{bmatrix}$$

The same LSO formula can be used for estimating any trend that is linear over the parameters, for example, polynomial trends, of course by adding more parameters

12.3 Seasonality Removal

Having an SP with seasonality, we can decompose it as:

$$y(t) = \tilde{y}(t) + s(t)$$

where s is a function such that for some $k \in \mathbb{Z}$ and some fixed T :

$$s(t) = s(t + kT)$$

Remark.

Usually T is given, but it could be identified in the spectrum, realizing in which part of the frequency w has a peak ($w = \frac{2\pi}{T}$)

Again, we can forecast with the same procedure seen in algorithm 3.

Now, we must estimate the period with the dataset. For this, we assume the dataset has M periods. Then, we can extract the SSP from the process with:

$$\hat{S}(t) = \frac{1}{M} \sum_{h=0}^{M-1} y(t + hT), \quad \forall t = 1, \dots, T$$

Note that it must be satisfied: $M \cdot T \leq N$.

Note that:

$$\begin{aligned}\hat{S}(t) &= \frac{1}{M} \sum_{h=0}^{M-1} y(t+hT) = \frac{1}{M} \sum_{h=0}^{M-1} (\tilde{y}(t+hT) + s(t+hT)) \\ &= \frac{1}{M} \sum_{h=0}^{M-1} \tilde{y}(t+hT) + \frac{1}{M} \sum_{h=0}^{M-1} s(t+hT)\end{aligned}$$

And taking the expected value:

$$\begin{aligned}E[\hat{S}(t)] &= E\left[\underbrace{\frac{1}{M} \sum_{h=0}^{M-1} \tilde{y}(t+hT)}_{\text{if } M \text{ is large, } =0}\right] + E\left[\underbrace{\frac{1}{M} \sum_{h=0}^{M-1} s(t+hT)}_{=s(t) \text{ (Since } s(t)=s(t+hT))}\right] \\ &= s(t)\end{aligned}$$

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